Computer Vision — 21CSE454T

Unit 4: Vision & Motion

Topic 1: Methods for 3D Vision | SLO1

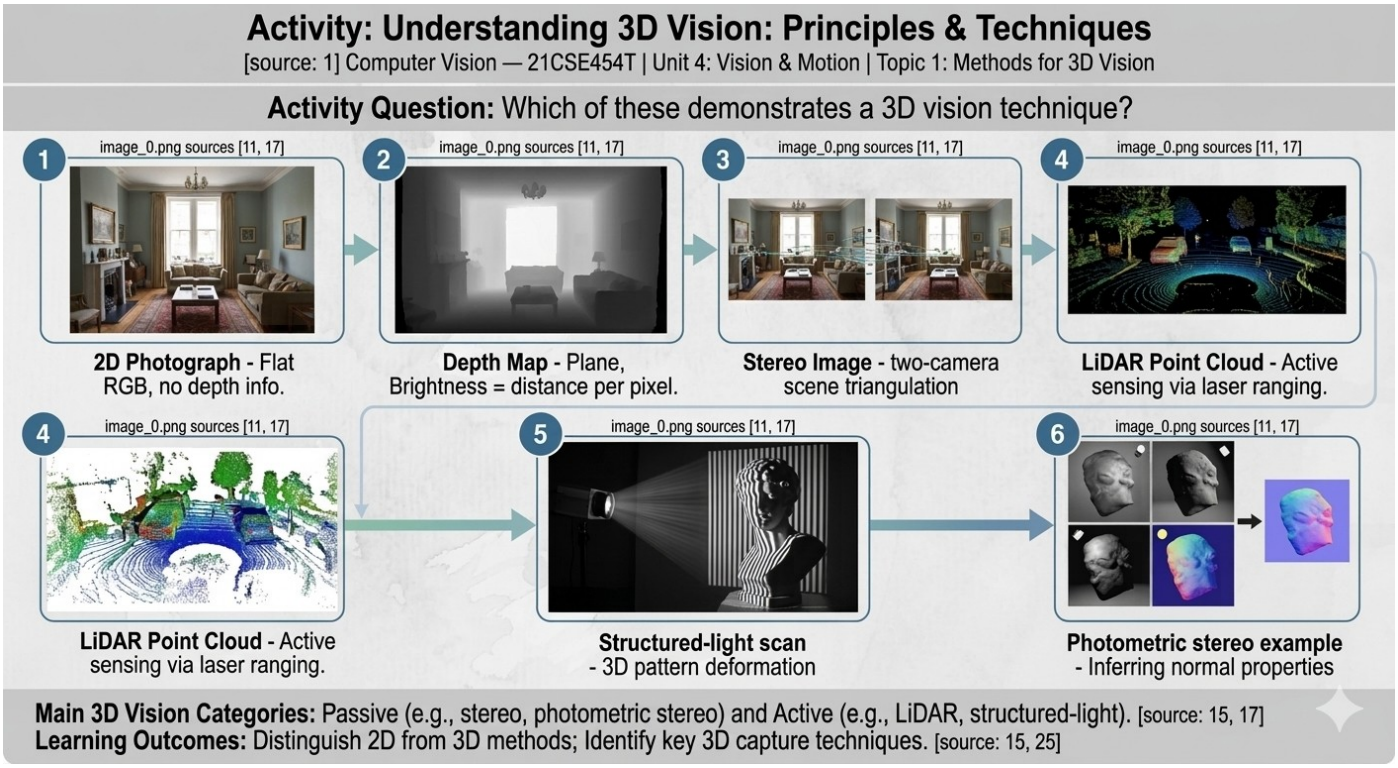
Course	Computer Vision — 21CSE454T
Unit	Unit 4: Vision & Motion
Topic	Topic 1: Methods for 3D Vision
SLO1	Explain the principles and techniques behind 3D vision methods.
Session Title	Understanding 3D Vision: Principles & Techniques
Duration	10 Minutes Activity
Response Method	YES / NO cards | Thumbs 👍 / 👎 | Labelled flashcards

Activity Question

> ***"Which of these images / diagrams represents a 3D vision technique in action?"***
>
> Answer: Images 2, 3, 4, 5, and 6 — YES. Image 1 — NO, because it is a flat 2D RGB photograph with no depth information.

Activity Overview

Students examine 6 image patches — point clouds, depth maps, stereo pairs, and 2D photographs. They vote on whether each image demonstrates a 3D vision technique, then discuss what distinguishes 3D from 2D vision.



Instructor Instructions

1. Display 6 images: a 2D flat photo, a depth map, a stereo image pair, a LiDAR point cloud, a structured-light scan, and a shaded surface from photometric stereo.
2. For each image, ask: 'Does this demonstrate a 3D vision technique?'
3. Students raise YES / NO cards or show thumbs up / down.
4. After each vote, briefly explain the technique shown and why it is (or isn't) 3D vision.
5. Close by listing the main 3D vision categories: passive (stereo, shape-from-X) and active (structured light, ToF, LiDAR).

Step-by-Step Activity Flow

Step	Action	Details / Expected Student Response
1	**2D Photograph**	Flat RGB photo of a room — no depth info. Expected: NO. Discuss: cameras project 3D onto 2D, losing depth.
2	**Depth Map**	Grayscale image where brightness = distance. Expected: YES. Explains per-pixel depth encoding.
3	**Stereo Image Pair**	Two offset views of same scene. Expected: YES. Introduces binocular disparity and triangulation.
4	**LiDAR Point Cloud**	Coloured 3D point scatter of street scene. Expected: YES. Active sensing via laser ranging.
5	**Structured-Light Scan**	Projected stripe pattern on object. Expected: YES. Deformation of pattern reveals depth.
6	**Photometric Stereo**	4 images of same surface under different lights. Expected: YES. Normals recovered from shading.

Student Response Method

After each image or scenario is displayed, students respond using one of:

- Raising YES card (green) or NO card (red)
- Showing thumbs up 👍 for YES or thumbs down 👎 for NO
- Holding up labelled category cards (e.g., ORTHO / PERSP / OBLIQUE)

The instructor acknowledges the class response, reveals the correct answer, and provides a 1–2 sentence explanation before moving to the next item.

Expected Learning Outcomes

#	Learning Outcome
1	Distinguish between 2D image analysis and 3D vision methods.
2	Name at least four techniques used to capture or infer 3D structure.
3	Classify 3D vision methods as active (sensor-emitting) or passive (light-only).
4	Explain why a standard photograph is insufficient for 3D reconstruction.